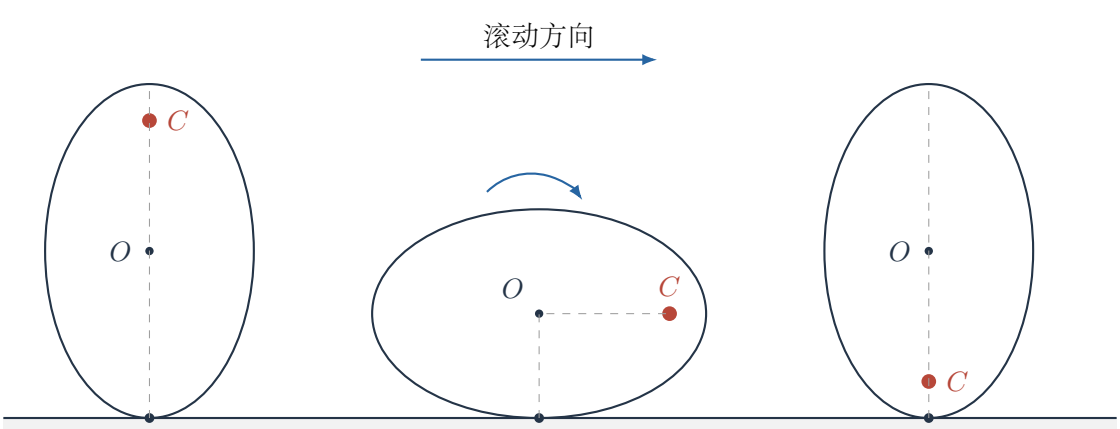
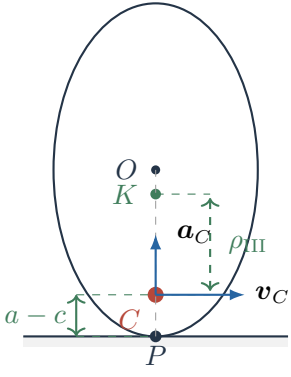
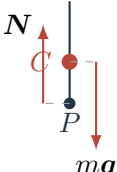


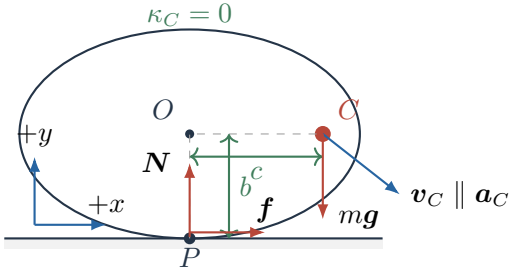
P 为瞬时接触点



状态 III 运动学



状态 III 受力



状态 II